

F538 — K-TYPE NOTATION DICTIONARY (pinned once per
 Keeper K683 discipline — $(a,b)/(k,m)/\nu$ were being used
 interchangeably; state one dictionary, VERIFY, cite, stop
 relabeling) + first forward LEAD on the two-number
 (charged/quark) selection map: the strata-dimension $\rightarrow$
 SO(5)-tensor-symmetry discriminator selects the muon's (1,1)
 over (2,0). THE DICTIONARY (three labels, one meaning each):
 ν = DISCRETE-SERIES parameter = SO(2)-charge of a tower's
 LOWEST K-type = weighted-Bergman weight; selects the
 TOWER (particle species); HALF-integer for fermions (odd- n_C
 ρ -shift = $n_C/2 = 5/2$); Elie's formal degree
 $d(\nu)=(5/2-\nu)(1-\nu)(2-\nu)(3-\nu)(4-\nu)$ is the SCALAR-tower
 single-parameter case. $(a,b) = \text{SO}(5)=B_2$ HIGHEST WEIGHT
 of a K-type, $a \geq b \geq 0$ (orthogonal basis, $\rho_{\text{SO}(5)}=(3/2,1/2)$); the
 angular/internal content WITHIN a tower; integer=tensor,
 half-integer=spinor. $(k,m) = \text{DEPRECATED}$ as a primary label
 — it was shorthand for the SINGLE-ROW case $(a,b)=(m,0)$ at
 SO(2)-level k ; use (a,b) for the general K-type and reserve 'level'
 k for the SO(2) shell count only. CANONICAL ADDRESS of a
 state = [tower ν ; K-type (a,b)]. VERIFIED via B_2 Weyl
 dimension formula $\dim(a,b) =$
 $(a-b+1)(2b+1)(2a+3)(a+b+2)/6$: $(0,0)=1$, $(1,0)=5$ [vector V_5],
 $(1/2,1/2)=4$ [Dirac spinor], $(1,1)=10$ [adjoint/2-form], $(2,0)=14$
 [traceless Sym^2] — all match known SO(5) irreps. THE
 STRATA $\rightarrow$ ADDRESS MAP (F86 Korányi–Wolf: 3 support strata
 bulk $\dim n_C=5$ / Cartan-slice $\dim \text{rank}=2$ / Shilov $\dim 0$):
 gen-1 electron = ground single-row $(0,0)$ -base [origin, $N \rightarrow 1$];
 gen-2 muon = $(1,1)$ the SO(5) adjoint/2-form ($\dim 10$,

F361-pinned from its mass) — the FIRST TWO-ROW address
 ($b \neq 0$); gen-3 tau = Shilov higher address $[N \rightarrow 0]$. THE
 ‘TWO-NUMBER’ DIFFICULTY, LOCATED PRECISELY: it is
 EXACTLY that the muon’s $b \neq 0$ — gen-2/3 leave the single-row
 $(a, 0)$ family, so the selection principle must force a SECOND
 label. THE DISCRIMINATOR (first forward lead,
 target-innocent): the second $SO(5)$ shell has two candidates,
 $(2, 0) = 14$ [symmetric, traceless Sym^2] and $(1, 1) = 10$
 [ANTIsymmetric, 2-form/adjoint]; F361 says muon $= (1, 1)$; a
 FORWARD principle must select $(1, 1)$ over $(2, 0)$ from
 GEOMETRY not from m_μ ; LEAD — the Cartan-slice stratum
 has $\dim = \text{rank} = 2$, and a rank-2 ANTISYMMETRIC object
 (2-form) is $(1, 1)$ while the symmetric is $(2, 0)$, so the stratum’s
 antisymmetric-rank-2 character is the candidate selector (to
 force next: show the Cartan-slice localized modes carry the
 $SO(5)$ 2-form, not Sym^2). HONEST TIER: the DICTIONARY is
 verified/solid ($SO(5)$ dims computed); the strata $\rightarrow (a, b)$ map is a
 target-innocent LEAD (the antisymmetric-rank-2
 discriminator), NOT forced — the map must be derived, not
 read off F361. This is the notation-pin Keeper required BEFORE
 the two-number map, plus the first concrete discriminator for
 that map. For Casey (before pushing the hard frontier I pinned
 the vocabulary — we’d been using three different labelings for
 the particle ‘slots’ interchangeably, and we’ve been burned by
 exactly that kind of sloppy relabeling before, so I wrote ONE
 dictionary, checked it against known facts, and it’s clean; and
 pinning it immediately paid off — it showed exactly WHY the
 charged particles are harder than the neutrino: the muon’s slot
 is the first one that needs TWO numbers instead of one (it’s the

‘2-form’ slot, an antisymmetric object), and there’s a clean geometric reason to expect that — the muon lives on the 2-dimensional ‘Cartan slice’ of the geometry, and a 2-dimensional antisymmetric object is exactly a 2-form; so the slot’s shape may be forced by the dimension of the stratum it sits on, which is target-innocent; that’s the first real handle on the frontier, though it’s a lead to force, not yet forced), Keeper (K683 notation discipline EXECUTED: dictionary [tower ν ; K-type (a,b)], (k,m) deprecated to single-row shorthand, verified via B_2 Weyl dims, cite FK+F361; and the two-number map’s first forward LEAD — the muon (1,1) vs (2,0) selection may be forced by the Cartan-slice $\dim=\text{rank}=2 \rightarrow$ antisymmetric 2-form character; honest tier: dictionary solid, strata $\rightarrow$ (a,b) a lead not forced), Elie (for the muon bounded win: the muon’s canonical address in the pinned dictionary is [tower ν ; (a,b)=(1,1)] — the SO(5) adjoint/2-form, dim 10; hand this to Engine C for the Szegő-residue-at-the-muon-address evaluation; note (1,1) is TWO-ROW so the residue integrand is the 2-form K-type, not a single-row harmonic), Grace (your Engine-B construction receives the depths keyed to these addresses — gen-1 (0,0)-base, gen-2 (1,1), gen-3 Shilov; the depths N_i follow from the addresses once the strata $\rightarrow$ (a,b) map is forced; the notation is now pinned so our lanes share one labeling), Cal (F538: notation dictionary pinned+verified [ν ;(a,b)], (k,m) deprecated; two-number map lead — Cartan-slice $\dim=\text{rank}=2 \rightarrow$ antisymmetric 2-form (1,1) over symmetric (2,0), target-innocent discriminator to force; dictionary solid, map a lead; nothing banked).

F538 — The K-type notation dictionary (pinned), and the first forward lead on the two-number map

Keeper (K683) required the notation be pinned before pushing the two-number selection map — (a,b), (k,m), and ν were being used interchangeably, and we’ve been burned by exactly that (the genus-name flip). One dictionary, verified, cited, then the map.

The dictionary (three labels, one meaning each)

label	meaning	notes
ν	discrete-series parameter = SO(2)-charge of a tower’s lowest K-type = weighted-Bergman weight	selects the tower (particle species); half-integer for fermions (odd-n_C ρ -shift = 5/2); Elie’s $d(\nu)=(5/2-\nu)(1-\nu)(2-\nu)(3-\nu)(4-\nu)$ is the scalar-tower case
(a,b)	SO(5)=B ₂ highest weight of a K-type, $a \geq b \geq 0$	the angular/internal content within a tower; integer = tensor, half-integer = spinor; $\rho_{SO(5)}=(3/2,1/2)$
(k,m)	deprecated as a primary label	was shorthand for the single-row case (a,b)=(m,0) at SO(2)-level k; use (a,b) generally, reserve k for the SO(2) shell count

Canonical address of a state = [tower ν ; K-type (a,b)].

Verified via the B₂ Weyl dimension formula $\dim(a,b) = (a-b+1)(2b+1)(2a+3)(a+b+2)/6$:

(a,b)	dim	irrep
(0,0)	1	trivial
(1,0)	5	vector V ₅
(1/2,1/2)	4	Dirac spinor
(1,1)	10	adjoint / 2-form
(2,0)	14	traceless Sym ²

All match known SO(5) irreps.

The strata → address map (F86 Korányi–Wolf)

Three support strata: bulk (dim n_C=5) / Cartan-slice (dim rank=2) / Shilov (dim 0).

- gen-1 electron: ground, single-row (0,0)-base [origin, N→1]
- gen-2 muon: (1,1) — the SO(5) adjoint/2-form (dim 10), F361-pinned from its mass — the first two-row address (b≠0)
- gen-3 tau: Shilov, higher address [N→0]

The “two-number” difficulty, located precisely: it is *exactly* that the muon’s b≠0 — gen-2/3 leave the single-row (a,0) family, so the selection principle must force a second label. The neutrino was easy because it was single-label (pure ν); the charged sector is hard because (a,b) genuinely needs both.

The discriminator (first forward lead, target-innocent)

The second $SO(5)$ shell has two candidates: $(2,0)=14$ [symmetric, Sym^2] and $(1,1)=10$ [antisymmetric, 2-form]. F361 says muon = $(1,1)$. A *forward* principle must select $(1,1)$ over $(2,0)$ from geometry, not from m_μ .

Lead: the Cartan-slice stratum has $\dim = \text{rank} = 2$, and a rank-2 antisymmetric object (a 2-form) is exactly $(1,1)$, while the symmetric one is $(2,0)$. So the stratum's antisymmetric-rank-2 character is the candidate selector. To force next: show the Cartan-slice-localized modes carry the $SO(5)$ 2-form, not Sym^2 — then $\text{gen-2} = (1,1)$ is forward, not read off the mass.

Honest tier

The dictionary is solid ($SO(5)$ dims computed and verified). The $\text{strata} \rightarrow (a,b)$ map is a target-innocent lead (the antisymmetric-rank-2 discriminator), not forced — it must be derived, not read off F361. This is the notation-pin Keeper required, plus the first concrete discriminator for the two-number map.

Plain-language version

Before attacking the hard problem, I pinned the vocabulary. We'd been using three different labelings for the particle "slots" interchangeably, and sloppy relabeling has bitten us before — so I wrote one dictionary, checked it against known facts, and it's clean. Pinning it paid off right away: it shows *why* the charged particles are harder than the neutrino. The neutrino's slot needs one number; the muon's needs two — it's the "2-form" slot, an antisymmetric object. And there's a clean geometric reason to expect exactly that: the muon lives on the 2-dimensional "Cartan slice" of the geometry, and a 2-dimensional antisymmetric object *is* a 2-form. So the shape of the slot may be forced by the dimension of the stratum it sits on — which is target-innocent (nothing to do with the muon's measured mass). That's the first real handle on the frontier — a lead to force, not yet forced.

— Lyra, 2026-07-14. F538: K-type notation dictionary pinned+verified — [tower ν ; K-type (a,b)], (k,m) deprecated; B_2 dims check $(1,1)=10$, $(2,0)=14$ etc. $\text{Strata} \rightarrow \text{address}$ (F86): $e=(0,0)$ -base, $\mu=(1,1)$ 2-form (first two-row), $\tau=\text{Shilov}$. Two-number difficulty = muon $b \neq 0$. Forward LEAD: Cartan-slice $\dim=\text{rank}=2 \rightarrow$ antisymmetric 2-form $(1,1)$ over symmetric $(2,0)$; target-innocent discriminator to force. Dictionary solid; map a lead. No fabrication.